Chapter 5 Answers

Problem 1

(a): Consider ω_n, that is, the set of all x such that $r(n,x) \in T_0$. In the last chapter, we showed that T_0 and the relation $g(x) = y$ are both recursively enumerable. Hence $r(n,x) \in T_0$ iff $g(x) = x_0 \wedge nx_0 \in T_0$. Now, the relation $r(x,y) = z$ is recursively enumerable, as it is expressed by:

$$\exists x_0 (g(x) = x_0 \wedge yx_0 = z)$$

and thus the relation $r(n,x) \in T_0$ is expressed by the formula:

$$\exists z (r(n,x) = z \wedge z \in T_0)$$

(b): Suppose that A is a recursively enumerable set. Hence there is some predicate P of (U) such that $Hx \in T$ iff $x \in A$. If p is the Gödel number of P, $Px \in T$ iff $px_0 \in T_0$ iff $r(p,x) \in T_0$ iff $x \in \omega_p$, thus proving that $A = \omega_p$.

Exercises 1-3

I could not solve any of these.

Problem 2

Consider a recursively enumerable relation $R(x,y)$. Hence, in (U), it is represented by some predicate H, with a Gödel number of h. Hence $R(x,y)$ holds iff $hxcy \in T_0$ for c the Gödel number of the comma. Take $t(y)$ to be hy_0c, which is recursive as $g(x) = x_0$ is recursively enumerable. Hence $x \in \omega_{t(i)}$ iff $r(hi_0c,x) \in T_0$ iff $hi_0cx \in T_0$ iff $R(i,x)$.

Note: The above is essentially the answer that Smullyan gave, but it is incorrect. It proves that $x \in \omega_{t(i)}$ iff $R(i,x)$, but that was not the theorem! It was to prove that $x \in \omega_{t(i)}$ iff $R(x,i)$. Below is a correct proof:

We suppose that $R(x,y)$ is recursively enumerable. Let $R_1(x,y)$ be the relation $R(y,x)$. Hence $R_1(x,y)$ is recursively enumerable. Thus $R(x,y)$ is represented in (U) by some predicate H with Gödel number h, and $R_1(x,y)$ is represented in (U) by some predicate P with Gödel number p. Hence

$$px_0cy_0 \in T_0 \text{ iff } hy_0cx_0 \in T_0$$

for x_0, y_0, c the Gödel numbers for x, y, and the comma, respectively. Let $t(y)$ be the relation py_0c. Thus $x \in \omega_{t(i)}$ iff $r(pi_0c, x) \in T_0$ iff $pi_0cx_0 \in T_0$ iff $hx_0ci_0 \in T_0$ iff $R(x,i)$, meaning $x \in \omega_{t(i)}$ iff $R(x,i)$, thus completing the proof.

Problem 3

Consider some recursively enumerable relation $R(x_1, ..., x_m, y_1, ..., y_n)$. Hence there is a recursively enumerable relation $M(x,y)$ such that $R(x_1, ..., x_m, y_1, ..., y_n)$ iff $M(\delta_m(x_1, ..., x_m), \delta_n(y_1, ..., y_n))$. By the Simple Iteration Theorem, there is a function $t(i)$ such that $x \in \omega_{t(i)}$ iff $M(x,i)$. Hence, take $\phi(i_1, ..., i_n)$ to be $t(\delta_n(i_1, ..., i_n))$. Hence $M(\delta_m(x_1, ..., x_m), \delta_n(i_1, ..., i_n))$ iff $\delta_m(x_1, ..., x_m) \in \omega_{\phi(i_1,...,i_n)}$ iff $R_{\phi(i_1,...,i_n)}(x_1, ..., x_m)$. This is true as we recall that $R_i^n(x_1, ..., x_n)$ was defined to be $\delta_n(x_1, ..., x_n) \in \omega_i$.

The above proof handles the case where $n \geq 2$ and $m \geq 2$. The proof where either $n = 1$ and $m \geq 2$, or $m = 1$ and $n \geq 2$ (a case which Smullyan left out of his answer), is shown below.

Consider some recursively enumerable relation $R(x, y_1, ..., y_n)$. Take $M(x,y)$ to be the relation $R(x, K_1^n(y), K_2^n(y), ..., K_n^n(y))$ (where $K_i^n(x)$ is the recursive function defined in Chapter 4) which is recursively enumerable. Thus $M(x, \delta_n(y_1, y_2, ..., y_n))$ iff $R(x, y_1, y_2, ..., y_n)$. By the simple iteration theorem, there is some function $t(i)$ such that $x \in \omega_{t(i)}$ iff $M(x,i)$. Thus $x \in \omega_{t(\delta_n(y_1,y_2,...,y_n))}$ iff $M(x, \delta_n(y_1, y_2, ..., y_n))$ iff $R(x, y_1, y_2, ..., y_n)$, and $x \in \omega_{t(\delta_n(y_1,y_2,...,y_n))}$ is the same as saying $R_{t(\delta(y_1,y_2,...,y_n))}(x)$, as $m = 1$ (in $R(x_1, ..., x_m, y_1, ..., y_n)$). Thus $R(x, y_1, y_2, ..., y_n)$ iff $R_{t(\delta_n(y_1,y_2,...,y_n))}(x)$, thus proving that $t(\delta(y_1, y_2, ..., y_n))$ is the desired function $\phi(i_1, i_2, ..., i_n)$.

Now consider some recursively enumerable relation $R(x_1, x_2, ..., x_m, y)$. Take $M(x,y)$ to be the relation $R(K_1^m(x), K_2^m(x), ..., K_m^m(x), y)$ which is recursively enumerable. Thus $M(\delta_m(x_1, x_2, ..., x_m), y)$ iff $R(x_1, x_2, ..., x_m, y)$. By the simple iteration theorem, there is some function $t(i)$ such that $x \in \omega_{t(i)}$ iff $M(x,i)$. Thus $\delta_m(x_1, x_2, ..., x_m) \in \omega_{t(i)}$ iff $M(\delta_m(x_1, x_2, ..., x_m), i)$ iff $R(x_1, x_2, ..., x_m, y)$,

and $\delta_m(x_1, x_2, ..., x_m) \in \omega_{t(i)}$ is defined to be equivalent to $R_{t(i)}(x_1, x_2, ..., x_m)$. Thus $R(x_1, x_2, ..., x_m, y)$ iff $R_{t(i)}(x_1, x_2, ..., x_m)$, proving that $t(i)$ is the desired function $\phi(i_1)$ in this case.

Problem 4

We consider the enumeration $\omega_1, \omega_2, ..., \omega_n, ...$ and any other recursively enumerable indexing $A_1, A_2, ..., A_n, ...$ Hence $x \in A_y$ is recursively enumerable, which means there is a recursive function $t(i)$ such that $x \in \omega_{t(i)}$ iff $x \in A_i$, thus completing the proof.

Problem 5

Define $S(x_1, ..., x_m, y, y_1, ..., y_n)$ iff $R_y(x_1, ..., x_m, y_1, ..., y_n)$, which is recursively enumerable because $R_y(x_1, ..., x_m, y_1, ..., y_n)$ is recursively enumerable by the enumeration theorem. Hence, by the iteration theorem, there exists some $\phi(i, i_1, ..., i_n)$ such that $R_{\phi(i,i_1,...,i_n)}(x_1, ..., x_m)$ iff $S(x_1, ..., x_m, y, y_1, ..., y_n)$ iff $R_i(x_1, ..., x_m, i_1, ..., i_n)$.

Problem 6

Let n be the number $t(m)$. Hence $R(x, n)$ iff $R_m(x, m)$ iff $x \in \omega_n$.

Note: To be clear, if $R(x, y)$ is recursively enumerable, then so is $R(x, t(y))$, as it is equivalent to the recursively enumerable relation:

$$\exists z(t(y) = z \wedge R(x, z))$$

By the enumeration theorem, there then exists a relation $R_m(x, y)$ such that $R_m(x, y)$ iff $R(x, t(y))$, as $R(x, t(y))$ is a relation between x and y.

Problem 7

(a): Consider some recursive function $f(x)$. Hence $f(x) = y$ is recursively enumerable, and thus the relation $R(x, y)$:

$$\exists z(f(y) = z \wedge x \in \omega_z)$$

is recursively enumerable. Thus by the weak recursion theorem, there is some number n such that $x \in \omega_n$ iff $R(x, n)$ iff $x \in \omega_{f(n)}$.

(b): 1: We can just take $f(x)$ to be $x + 1$ (which is recursive, as the relation $x + 1 = y$ is easily shown to be recursively enumerable), meaning, by part (a), there is some number n such that $\omega_n = \omega_{n+1}$.
3: This is a special case of 4, when $f(x) = x$.
4: For a recursive function $f(x)$, $x = f(y)$ is recursively enumerable (because $f(y) = x$ is recursively enumerable). Hence by Problem 5, there is some number n such that $x \in \omega_n$ iff $x = f(n)$, that is, $\omega_n = \{f(n)\}$.
5: Consider the relation $y \in \omega_x$, which is recursively enumerable. Hence, by Problem 5, there is some number n such that $x \in \omega_n$ iff $n \in \omega_x$.

Problem 8

(a): We know that $A \neq \mathbb{N}$ and $A \neq \emptyset$ and that A is a recursive set. Thus there is some number $a \in A$ and some number $b \notin A$. Let $f(x)$ be the function:

$$f(x) = \begin{cases} x \in A: & b \\ x \notin A: & a \end{cases}$$

$f(x) = y$ is equivalent to the relation $R(x, y)$:

$$(x \in A \wedge y = b) \vee (x \notin A \wedge y = a)$$

which is recursive as A is recursive. Thus $f(x)$ is recursive. Obviously, $x \in A$ iff $f(x) \notin A$.

(b): Consider the $f(x)$ of part (a). Because it is recursive, by Problem 7, there is some number n such that $\omega_{f(n)} = \omega_n$. Because $f(n) \in A$ iff $n \notin A$, either $f(n)$ or n are in A, but not both. But this means that A is not extensional, for otherwise both $f(n)$ and n would need to be in A.

Problem 9

(a): Take $\phi(i)$ to be the function $d(t(i))$. Thus $x \in \omega_{\phi(i)}$ iff $x \in \omega_{d(t(i))}$ iff $R_{t(i)}(x, t(i))$ iff $M(x, i, \phi(i))$, which is what we were required to prove.

Note to self: Here is why $M(x, z, d(y))$ is recursively enumerable. Because $M(x, y, z)$ is, then so is $M(x, y, d(z))$. Then let $M_1(x, y, z)$ be the relation $M(x, z, d(y))$ which is recursively enumerable. By the iteration theorem, there is some funciton $t(i)$ such that $R_{t(i)}(x, y)$ iff $M_1(x, y, i)$ iff $M(x, i, d(y))$.

(b): Take $\phi(i)$ to be the function $t(i,h)$. Thus $x \in \omega_{t(i,h)}$ iff $R_h(x,i,h)$ iff $M(x,i,t(i,h))$, which is what we were required to prove.

Note to self: Here is why $R_z(x,y,z)$ can be said to be a recursively enumerable relation. $R_x(y,z,z_1)$ is defined to be a recursively enumerable relation. Hence let $R_1(x,y,z)$ be the relation $R_x(y,z,z_1) \wedge x = z_1$, which is recursively enumerable, and is the same as saying $R_z(x,y,z)$.

Problem 10

Let $M(x,y,z)$ be $R_y(x,z)$. Then by the strong recursion theorem there exists some function $\phi(i)$ such that

$$\omega_{\phi(i)} = \{x : M(x,i,\phi(i))\} = \{x : R_i(x,\phi(i))\}$$

which proves Theorem 8 is a special case of the strong recursion theorem.

Problem 11

Suppose that $f(x)$ reduces A to B and $\phi(x)$ is a generative function for A, and $t(y)$ is an iterative function for the relation $f(x) \in \omega_y$. Hence $x \in A$ iff $f(x) \in B$, $x \in \omega_{t(y)}$ iff $f(x) \in \omega_y$, and $\phi(i) \in A$ iff $\phi(i) \in \omega_i$. Thus $f(\phi(t(x))) \in B$ iff $\phi(t(x)) \in A$ iff $\phi(t(x)) \in \omega_{t(x)}$ iff $f(\phi(t(x))) \in \omega_x$. That is, $f(\phi(t(x))) \in B$ iff $f(\phi(t(x))) \in \omega_x$. If we let $\phi(i)$ be the function $f(\phi(t(x)))$, this proves that $\phi(i) \in B$ iff $\phi(i) \in \omega_i$.

Problem 12

For every i, $x \in \omega_i$ iff $\delta(x,i) \in K$. Hence let $f(x)$ be the function $\delta(x,i)$. Thus $x \in \omega_i$ iff $f(x) \in K$, thus showing K is universal under the function $f(x)$.

Note: Smullyan must have made a typo when defining K. He clearly means that $x \in \omega_y$ iff $\delta(x,y) \in K$, but the definition was $\partial(x,y) \in K$, which is an undefined function.

Problem 13

The set C is recursively enumerable, as it is equivalent to the recursively enumerable relation $x \in \omega_x$. Now consider some universal set U. Hence C is reducible to U. Because $I(x)$ is a generative function for C, $f(I(t(x)))$ (as defined in Theorem 9^*) is a generative function for U by Theorem 9^*.

Problem 14

We can show that Post's set K is uniformly universal. We have shown that it is universal under the function $\delta(x, i)$ for every ω_i. Hence K is uniformly universal under the recursive function $\delta(x, y)$, as $x \in \omega_i$ iff $\delta(x, i) \in K$. Now consider any universal set U. Hence, for some recursive function $f(x)$, $x \in K$ iff $f(x) \in U$. Take $f(x, y)$ to be the function $f(\delta(x, y))$. Hence, for all i, $f(\delta(x, i)) \in U$ iff $\delta(x, i) \in K$ iff $x \in \omega_i$, thus proving that U is uniformly universal under the recursive function $f(\delta(x, y))$.

Exercise 4

Suppose that A is uniformly universal under $f(x, y)$ and that $t(y)$ is an iterative function for the relation $f(x, x) \in \omega_y$. Thus, for all i, $x \in \omega_i$ iff $f(x, i) \in A$, and $x \in \omega_{t(y)}$ iff $f(x, x) \in \omega_y$. Hence $f(t(x), t(x)) \in A$ iff $t(x) \in \omega_{t(x)}$ iff $f(t(x), t(x)) \in \omega_x$, thus proving $f(t(x), t(x))$ is a generative function for A.

Problem 15

(1): We assume that B is a generative set under $\phi(x)$, meaning $\phi(x) \in B$ iff $\phi(x) \in \omega_x$. Thus if $\omega_i = \mathbb{N}$ then, by definition, $\phi(i) \in \omega_i$ (as all positive integers are in ω_i) and thus $\phi(i) \in B$.

(2): If $\omega_i = \emptyset$, then $\phi(i) \notin B$. For if it was, then $\phi(i) \in \omega_i$. But this would be contrary to the assumption that $\omega_i = \emptyset$.

Problem 16

Suppose that A is a recursively enumerable set. Define $M(x, y)$ iff $y \in A$. Because $M(x, y)$ is recursively enumerable, by the iteration theorem, there is some recursive function $t(y)$ such that for all i, $\omega_{t(i)} = \{x : M(x, i)\}$.

(1): Suppose $i \in A$. Hence $x \in \omega_{t(i)}$ iff $M(x, i)$ iff $i \in A$, which shows that, because $i \in A$, $x \in \omega_{t(i)}$ for every x.

(2): Suppose $i \notin A$. By part (1) of this problem, we know that $x \in \omega_{t(i)}$ iff $i \in A$. Hence, because $i \notin A$, $x \notin \omega_{t(i)}$ for every x, thus meaning $\omega_{t(i)} = \emptyset$.

Problem 17

Consider some generative set B. By Problem 15, we know that, if B is generative under the recursive function $\phi(x)$, then, for every i:
(1) If $\omega_i = \mathbb{N}$ then $\phi(i) \in B$.
(2) If $\omega_i = \emptyset$ then $\phi(i) \notin B$.
Now consider any recursively enumerable set A. Hence, by Problem 16, there is a recursive function $t(y)$ such that for every number i:
(1) If $i \in A$, then $\omega_{t(i)} = \mathbb{N}$;
(2) If $i \notin A$, then $\omega_{t(i)} = \emptyset$.
Now, consider the function $\phi(t(i))$. If $i \in A$, then $\omega_{t(i)} = \mathbb{N}$ and thus $\phi(t(i)) \in B$. On the other hand, if $\phi(t(i)) \in B$, then $\phi(t(i)) \in \omega_{t(i)}$. If $i \notin A$, then $\omega_{t(i)} = \emptyset$, which would be contrary to the fact that $\phi(t(i)) \in B$ (by (2) of Problem 15). Thus $i \in A$. We have therefore shown that $i \in A$ iff $\phi(t(i)) \in B$, which shows that A is reducible to B under the function $\phi(t(i))$, and therefore B is universal.

Problem 18

Assume C_1, that is: There is a recursive function $\phi(x)$ such that for every i such that ω_i is disjoint from A and such that ω_i contains at most one member, the number $\phi(i)$ lies outside both A and ω_i.

(1): Suppose $\omega_i = \emptyset$. Suppose $\phi(i) \in A$. Hence ω_i is disjoint from A, and ω_i contains at most one member. Thus $\phi(i)$ lies outside both A and ω_i, contrary to the assumption $\phi(i) \in A$. Hence $\phi(i) \notin A$.

(2): Suppose $\omega_i = \{\phi(i)\}$. Suppose that $\phi(i) \notin A$. Then A and ω_i are disjoint. But because ω_i has only one member, $\phi(i)$ lies outside both A and ω_i,

contrary to the assumption that $\phi(i) \in \omega_i$. Thus $\phi(i) \in A$.

Problem 19

Consider some recursive function $\phi(x)$. Let $M(x, y, z)$ be the recursively enumerable relation $x \in \omega_y \wedge x = \phi(z)$. By the strong recursion theorem, there is some recursive function $t(i)$ such that $x \in \omega_{t(i)}$ iff $M(x, i, \phi(i))$ iff $x \in \omega_i \wedge x = \phi(t(i))$ iff $x \in \omega_i \cap \{\phi(t(i))\}$, which is the same as saying $\omega_{t(i)} = \omega_i \cap \{\phi(t(i))\}$. Thus $t(i)$ is the desired recursive function.

Problem 20

Suppose that $\phi(x)$ is a weakly co-productive function for A. Let $t(y)$ be the recursive function related to $\phi(x)$ as defined in Problem 19. Thus if $\phi(t(i)) \in \omega_{t(i)}$, then $\phi(t(i)) \in \omega_i \cap \{\phi(t(i))\}$, which is the same as saying $\omega_{t(i)} = \{\phi(t(i))\}$, and thus $\phi(t(i)) \in A$ by C_2 (2). Now suppose that $\phi(t(i)) \in A$. If $\phi(t(i)) \notin \omega_{t(i)}$, then we know that $\phi(t(i)) \notin \omega_i \cap \{\phi(t(i))\}$ and $\omega_{t(i)} = \omega_i \cap \{\phi(t(i))\}$, thus meaning $\omega_{t(i)} = \emptyset$, and thus $\phi(t(i)) \notin A$ by C_2 (1). Thus $\phi(t(i)) \in \omega_{t(i)}$. Hence we have shown that $\phi(t(i)) \in A$ iff $\phi(t(i)) \in \omega_{t(i)}$, thus proving that $\phi(t(i))$ is a generative function for A.

I will note that Smullyan's Theorem 14 seems incorrect. The proof was for co-productive sets, not productive ones, and I don't think its provable that every productive set is a co-productive one.

Direct Proof that Weak Co-productivity Implies Universality

Suppose that B is weakly co-productive under $\phi(x)$ and that A is a set that we wish to reduce to B (that is, a recursively enumerable set). Let $M(x, y, z)$ be the recursively enumerable relation

$$y \in A \wedge x = \phi(z)$$

By the strong recursion theorem, there is some recursive function $t(i)$ such that

$$(1)\ x \in \omega_{t(i)} \text{ iff } M(x, i, t(i)) \text{ iff } i \in A \wedge x = \phi(t(i))$$

Now suppose that $i \in A$. There are then two cases. In the first case, $x = \phi(t(i))$. Thus by (1), $x \in \omega_{t(i)}$. In the second case, $x \neq \phi(t(i))$. Again by (1), this would mean that $x \notin \omega_{t(i)}$. Thus $x \in \omega_{t(i)}$ iff $x = \phi(t(i))$, and therefore $\omega_{t(i)} = \{\phi(t(i))\}$. On the other hand, suppose that $i \notin A$. Thus, by (1), $x \notin \omega_{t(i)}$, and because this holds for all x, $\omega_{t(i)} = \emptyset$ in this case.

We now have the following. If $i \in A$, then by the above result, $\omega_{t(i)} = \{\phi(t(i))\}$. Thus $\phi(t(i)) \in B$ by C_2 (2). If $\phi(t(i)) \in B$, then $\omega_{t(i)} \neq \emptyset$ by C_2 (1). If $i \notin A$, then $\omega_{t(i)} = \emptyset$ by the above result, which is a contradiction. Hence $i \in A$. Thus, we have shown that $i \in A$ iff $\phi(t(i)) \in B$, which proves that $\phi(t(i))$ reduces A to B, thus showing that B is universal.

Note: When Smullyan said $\omega_{t(y)} = \{\phi(y)\}$ in the explanation of this problem, he actually meant $\omega_{t(y)} = \{\phi(t(y))\}$.

Exercise 5

To me it appears that for every sets A and B such that A is recursively isomorphic to B, then both A and B are recursively enumerable. For let $\phi(x)$ be the recursive isomorphism from A to B. Thus for every $x \in A$, $\phi(x) = y$ for some $y \in B$, and for every $y \in B$, there is one and only one number x such that $f(x) = y$. So the relation

$$\exists y(\phi(x) = y)$$

is a recursive relation for A. Because B is recursively isomoprhic to A under $\phi^{-1}(x)$, it also follows that

$$\exists y(\phi^{-1}(x) = y)$$

is a recursive relation for B. Thus neither A nor B can be generative, co-productive (which implies universality) or universal, as the Bonus Question shows that no generative or universal sets can be recursive.

Now suppose that A is productive. Hence there is some recursive function $\phi(i)$ such that, if ω_i is a subset of A, then $\phi(i) \in A - \omega_i$. Now let $f(x)$ be the recursive isomorphism from A to B. Hence $\phi(i) \in A$ iff $f(\phi(i)) \in B$. Thus if ω_i is a subset of B, then $f(\phi(i)) \in B$ iff $\phi(i) \in A$ and thus $\phi(i) \in A - \omega_i$ meaning $\phi(i) \notin \omega_i$ and therefore $\phi(i) \in B - \omega_i$, thus meaning $f(\phi(i))$ is a productive function for B.

Bonus Question

Consider some universal set A, and suppose that it was recursive. Hence $\bar{A}$ would be recursively enumerable. We have shown that every universal set is generative. Thus there is some recursive function $\phi(i)$ such that $\phi(i) \in A$ iff $\phi(i) \in \omega_i$. Because $\bar{A}$ is recursively enumerable, it has an index n. Thus $\phi(n) \in A$ iff $\phi(n) \in \omega_n$ iff $\phi(n) \in \bar{A}$, which is impossible. Hence there can be no recursive universal set (and no recursive generative set).